

Project Approach

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1. Scientific and Technical Approach

1.1. Observation

Within NHFT's Business Intelligence (BI) and Performance Improvement teams, the BI Service has observed that demand and capacity planning across the Trust's 125+ services are often reactive. Teams rely on spreadsheets or manual forecasting on an ad-hoc basis, which can lead to inconsistent predictions and delays in identifying service pressures. I noticed that while some services maintain detailed waiting list reports, others lack predictive insight into future workload, resulting in inefficiencies and avoidable breaches.

This observation led me to question whether a standardised, data-driven approach could be developed to model demand and capacity in a consistent and scalable way.

1.2. Problem

The core problem is:

How can NHFT accurately forecast service demand and optimise capacity allocation using existing data sources, ensuring consistency, fairness, and operational efficiency across multiple service types?

Breaking this into sub-questions:

- Can forecasting models predict patient demand more accurately than current manual methods?
- How can we model staffing and session capacity to minimise 18-week breaches and underutilisation?
- Can a single framework be configured to work across diverse NHFT services?

These questions are specific enough to test and measurable in terms of model performance and operational outcomes.

1.3. Background Information

Before developing the framework for this project, I reviewed existing work carried out within NHFT's BI Service. Previous work on demand and capacity modelling were highly specific, ad-hoc and manual. The findings echoed a broader pattern observed across the NHFT, where many available Demand & Capacity models focus on single settings, produce static calculations, or require significant manual input.

I then explored national resources, including NHS England's Service-Level Demand and Capacity Models [1], which offer structured approaches but are primarily designed for acute hospital settings rather than community or mental health services. Similar patterns were found when reviewing the Health Economics Unit's Demand and Capacity Modelling Service [2], which again emphasises acute care pathways. I also examined the NHS AI Lab Skunkworks case study [3] on improving hospital bed allocation using AI at Kettering General Hospital. While this project demonstrates interesting use of machine learning and simulation for inpatient flow, it is tailored to acute inpatient activity and does not translate directly to NHFT's diverse service mix. These insights reinforced the gap identified internally: existing models are useful but narrowly scoped, and do not offer a scalable or configurable solution suitable for a trust with over 200 community, mental health, and specialist services.

Consultations with the BI Manager and Performance Improvement Manager further shaped the direction of the project. They highlighted the inconsistency of current data definitions, reliance on manual forecasting methods, and the limitations of existing Power BI-based outputs. Bringing

together internal insights and national models provide a strong foundation for selecting appropriate modelling techniques and designing a more flexible architecture suited to NHFT's service landscape.

1.4. Hypothesis

Based on the problem and research, my working hypothesis is:

If a demand and capacity modelling platform is developed using automated data pipelines, statistical time-series forecasting, and structured capacity modelling techniques, then the resulting framework will produce more reliable, consistent, and interpretable demand forecasts than current manual or ad-hoc methods used within the Trust.

Predicted outcomes include:

- **Improved forecasting reliability** through ETS model selection using AIC, rather than relying on manual judgement or ad-hoc spreadsheet methods.
- **Greater modelling consistency**, achieved through standardised data preparation and reusable modelling components.
- **Improved interpretability and transparency**, allowing service managers or analysts to understand demand patterns more clearly than with existing manual spreadsheets.
- **Fairer and more consistent demand benchmarking**, achieved by recalculating service-line-specific targets using a configurable percentile within the currently filtered dataset, rather than applying a single global assumption across services.

This hypothesis will be tested through pilot implementation and comparison of model results against historical and real-world outcomes.

1.5. Testing

The project will follow a controlled testing approach using selected pilot services (e.g., Adult Dietetics, Community Nursing, MSK and CAMHS Outpatients).

Control Group: Current manual or spreadsheet-based forecasting and planning processes.

Test Group: The new Demand and Capacity Modelling Platform, which generates automated monthly forecasts and percentile-based demand benchmarking outputs. I will:

- Use historical monthly demand data to train and fit the time-series forecasting models, with ETS / Holt-Winters as the dashboard default.
- Select the ETS specification using AIC-based model comparison rather than manual parameter choice.
- Recalculate service-line-specific demand targets using the configurable percentile framework within the currently filtered dataset.
- Compare the resulting forecast outputs and derived demand metrics across pilot services and time windows to check for consistency and interpretability.

Key measurable variables include forecast stability, consistency of service-line targets, demand ratio, sustainable caseload, and 18-week breach trends.

1.6. Analysis

Once the forecasting and dashboard outputs have been generated, the results will be analysed to evaluate the model's usefulness for operational planning. The forecasting component uses **AIC-based model selection** to compare candidate ETS specifications and choose the most suitable model for each series.

The analysis will focus on:

- Comparing forecast outputs across services and time windows to assess consistency and interpretability.
- Reviewing derived demand metrics such as Demand Ratio and Sustainable Caseload, which are recalculated using a configurable percentile-based target framework.
- Examining whether service-line-specific targets remain balanced when the filtered dataset changes, supporting fairer comparison across services.
- Using Dash visualisations to present forecast versus actual demand, demand benchmarking outputs, and capacity-related trend patterns.
- Reflecting on the robustness, interpretability, and scalability of the approach, particularly in relation to interactive dashboard use and changing service filters.

1.7. Presentation

Findings will be shared through:

- A **dashboard demonstration** to NHFT stakeholders showing forecast accuracy and scenario insights.
- A **written project report and presentation** for ZDAT3001 submission summarising the methodology, results, and organisational benefits.
- **Internal feedback sessions** with the BI and Performance Improvement teams to evaluate usability and identify next steps for wider rollout.

If results indicate further refinement is needed, I will revisit the hypothesis and model configuration to improve performance before full implementation.

2. Software Development Method

Although the project follows a largely linear (Waterfall like) sequence, moving from documentation and governance to data access, to modelling, and finally to visualisation, it is managed using Microsoft Planner in a Kanban-style workflow. This combination gives the project both the structure it needs and the flexibility required for technical refinement. Kanban is the most suitable methodology because:

- It allows flexible task management across multiple workstreams (data engineering, modelling, and visualisation).
- Progress can be visually tracked using Kanban boards (aligned with NHFT BI's existing workflow).
- Tasks can be prioritised dynamically as stakeholder feedback and data issues arise.
- It supports incremental delivery, where functional components (e.g., forecasting module, simulation engine, dashboard) are completed and demonstrated continuously rather than waiting for final integration.
- It encourages continuous improvement and collaboration across BI and Performance Improvement teams.

This approach aligns with NHFT's culture of iterative improvement, allowing progress to be monitored transparently while maintaining flexibility for mid-project refinements.

Method	Strengths	Limitations	Fit for This Project
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Waterfall	Clear, linear, milestone-driven; good for fixed requirements.	Inflexible; difficult to adjust if data issues or new insights arise.	Useful for overall high-level milestone structure, but too rigid for modelling and analytical iteration.
Agile (Scrum)	Sprint-based, collaborative, regular checkpoints.	Heavy ceremony overhead; requires multi-person team structure.	Helpful conceptually but not ideal for a single-developer project.
Kanban	Highly flexible; visual task flow; suitable for continuous delivery and parallel workstreams; supports dynamic reprioritisation.	Requires strong self-discipline to maintain WIP limits.	Best fit: aligns with BI working practices, supports modelling iteration, and integrates well with Microsoft Planner.

3. Design and Structure of the Project

The project follows a linear high-level structure, but each stage contains iterative sub-processes. The overarching pathway is:

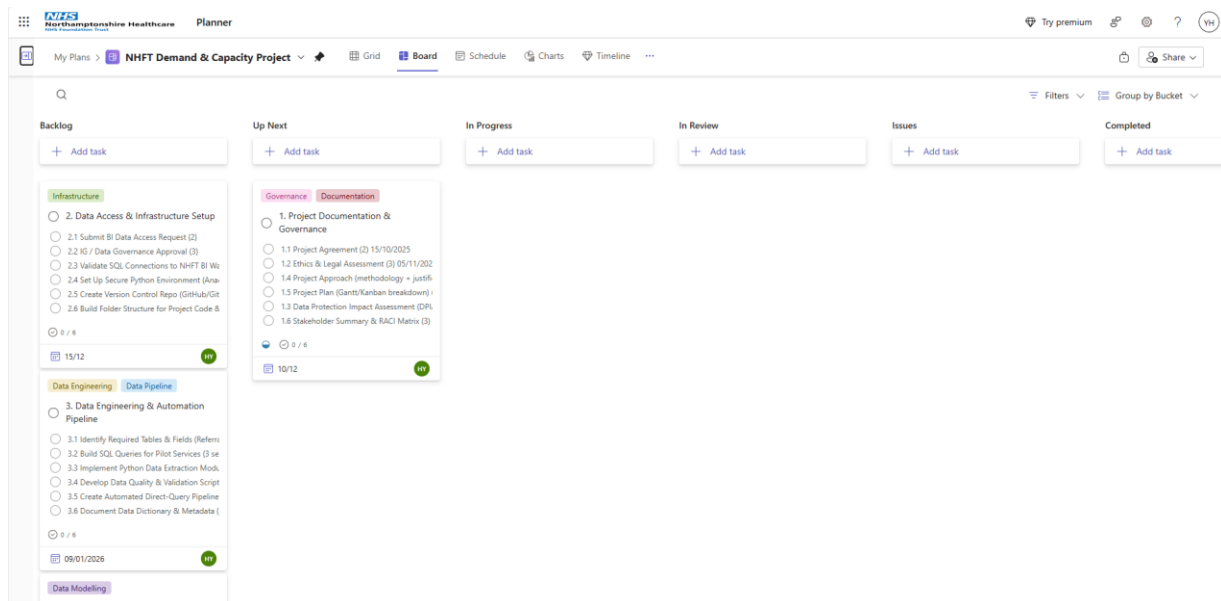
1. **Project Documentation & Governance** ensuring the project is defined, scoped, and compliant with NHFT and university requirements.
2. **Data Access & Infrastructure Setup** where the necessary SQL BI connections, secure environments, and governance approvals are established.
3. **Data Engineering & Automation Pipeline** focuses on building a direct-query pipeline that retrieves and validates data without the need for local extraction or storage.
4. **Forecasting Module (MVP)**, develops time-series models to predict future service demand
5. **Capacity & Simulation Module** explores the potential to incorporate queueing theory or discrete-event simulation to estimate system pressures and operational constraints.
6. **Dash / Power BI Application**, which transforms analytical outputs into an interactive visual tool for service managers, enabling them to explore forecasts and model different “what-if” scenarios.
7. **Testing, Validation & Refinement** ensures that the solution is robust, interpretable.
8. **Documentation, Logbook & Final Presentation Prep** packages the project for both organisational use and academic assessment.

These stages correspond directly with the project timeline below, and each stage is represented as an overarching task in Microsoft Planner. Subtasks allow detailed tracking, while the Planner columns support real-time progress monitoring. The Planner will include the following columns:

- **Backlog:** All planned and future tasks, grouped by overarching subjects.
- **Up Next:** Prioritised work items prepared for immediate action.
- **In Progress:** Tasks actively being developed or implemented.
- **In Review:** Work undergoing testing, validation, or stakeholder feedback.
- **Issues:** Any blocked items requiring clarification, rework, or risk mitigation.
- **Completed:** Tasks that meet acceptance criteria and are formally signed off.

Each overarching task (e.g., *Data Access*, *Forecasting Module*, *Simulation Engine*, *Dash App*) will contain subtasks organised using Microsoft Planner checklists.

To support planning and workload management, Fibonacci effort estimates (1, 2, 3, 5, 8, 13...) will be assigned to subtasks. This helps approximate complexity and time investment more realistically than fixed time estimates.



4. Risks and Quality

Several risks are associated with both the analytical and technical aspects of the project. One key risk is data accessibility, delays in obtaining SQL BI connections or governance approval could slow early progress. To mitigate this, the project front-loads all governance, DPIA, and access requests within the first project stage. A second risk relates to data quality and consistency, especially given the variation in data completeness and structure across NHFT services. Automated validation scripts, profiling checks, and logging will be implemented within the data pipeline to maintain data integrity and highlight anomalies early.

There are also analytical risks, particularly around forecasting accuracy and model generalisability. Time-series behaviour may differ significantly between services, which could result in inconsistent performance if models are not evaluated carefully. To address this, the project will adopt a rigorous evaluation process using metrics such as MAE and RMSE, alongside comparison of forecast error variance across services. Sensitivity analysis and fairness checks will also be used to ensure that the modelling framework does not over-perform on certain service types while underperforming on others.

Another risk relates to the more advanced components of the project, such as queueing theory and simulation, which are areas in which I am less experienced. This uncertainty will be mitigated through targeted academic guidance, reviewing published modelling approaches, and testing early prototypes on smaller problems before scaling to full pilot services. From a software quality perspective, risks include system instability, integration failures, or incorrect dashboard outputs. These will be managed through version control, modular code structure, incremental testing, and regular stakeholder reviews.

Quality assurance is woven throughout the project. Each stage includes validation activities from data verification to residual checks on forecasting models, to user testing on the dashboard interface. Documentation is maintained continuously through OneNote, the Git repository, and technical specifications, ensuring that all design decisions are explained and reproducible. This

commitment to iterative testing, transparency, and validation ensures the final solution is reliable, interpretable, and aligned with NHFT operational requirements.

Bibliography

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[3] NHS England, 'Improving hospital bed allocation using AI', NHS Transformation Directorate. Accessed: Dec. 27, 2024. [Online]. Available: <https://transform.england.nhs.uk/ai-lab/explore-all-resources/develop-ai/improving-hospital-bed-allocation-using-ai/>